

PAPER · VERSION 1.5A · APRIL 2026 · MPL-2.0

DOWNLOADS · [.docx](#)[← Back to dashboard](#)

The Borovoy Coefficient

A Civil-Liberties Measure of Legislative Alignment

Michael Bryant *Founder, justack.ai Former General Counsel and Executive Director, Canadian Civil Liberties Association (2018-2022)*

In honour of Alfred Alan Borovoy, O.C. (1932-2015), General Counsel of the Canadian Civil Liberties Association (1968-2009)

Version 1.5a — April 2026. Released under the Mozilla Public License 2.0 as a composite measurement instrument with companion scoring methodology and open dashboard at justack.ai/borovoy.

Abstract

Alan Borovoy spent fifty years cataloguing the phenomenon of legislative overreach: laws whose observed effects departed from their stated purposes, producing rights violations the drafters never advertised. His diagnostic method was an intent-effect analysis — asking whether a statute’s observed operation matched the purpose its preamble declared — and he carried it from the War Measures Act (1970) to Bill C-36 (2001) to the *Emergencies Act* invocation he did not live to see struck down (*Canadian Frontline Nurses v. Canada (Attorney General)*, 2024 FC 42).

This paper proposes that what Borovoy diagnosed is structurally similar to what machine-learning researchers call the *alignment problem*: the divergence between a specified

objective and its observed behaviour once the optimization process has done its work. The structural similarity is not merely metaphor. Hennessy & Goodhart (2023) have shown that penalized regressions under adversarial covariate manipulation generate Goodhart bias in a form admitting a common structural treatment for regulatory and machine-learning settings alike.¹ Canadian Charter jurisprudence, as recently crystallized in *R v Ndhlovu*, 2022 SCC 38, already reasons in this register — the Supreme Court struck mandatory and lifetime sex-offender registration on the ground that empirical evidence (75-80% of registered offenders never reoffend) showed the registry’s proxy had drifted from its investigative purpose. The same register is audible across the common-law bench. In *Johnson v United States*, 576 US 591 (2015), the US Supreme Court struck the Armed Career Criminal Act’s residual clause on the ground that its proxy for “violent felony” had produced empirical unpredictability courts could not cure; in *A v Secretary of State for the Home Department (Belmarsh)*, [2004] UKHL 56, the House of Lords issued a declaration of incompatibility against the post-9/11 detention regime for foreign-national terror suspects because its proxy for national-security risk was simultaneously over- and under-inclusive relative to the purpose asserted.

The Borovoy Coefficient is a 0-to-100 composite measure across five dimensions, each grounded in recognized constitutional doctrine (Canadian, US, and UK) and in the machine-learning alignment taxonomy. It scores legislation on the intent-effect gap. I apply it to fifteen representative laws as a prototype corpus. The paper concludes with five prospective drafting reforms — the Borovoy drafting principles — that would reduce Coefficients at source: sunset clauses, purpose-effect reporting, disparate-impact dashboards, narrowing presumptions, and independent oversight bodies with hard powers.

I. The Constitutional Alignment Problem

While reading Brian Christian’s *The Alignment Problem*,² a parallel from another discipline came to mind. I was taken back to a 2020 conversation with Cara Zwibel, a colleague at the Canadian Civil Liberties Association (CCLA), discussing Alan Borovoy, the long-time leader of CCLA, with whom Cara worked near his retirement. I have long observed that Borovoy defied traditional categorization, and his own writings resist doctrinal summation. His many books, articles, speeches and formal submissions to official bodies are pure legal discourse, advocacy, and he never sums up either his approach or that of the Canadian Civil Liberties

Association, which I led from 2018-22. He led it from 1968-2009. Zwibel credited Borovoy for saying that laws borne out of public angst tend to miss the mark, never hitting a bullseye, by matching purpose with effect. But attributing this to Borovoy is frustrated by his refusal to articulate a doctrine himself, notwithstanding a large corpus of Borovoy publications, legal facts, and CCLA's jurisprudential contributions. To put it into terms well known to science, Borovoy's lab made unprecedented contributions to Canadian jurisprudence, laws and public affairs. As he did put it, after some aggressive prodding by a skilled journalist, "our whole mandate is to change the law."³

Inspired by Borovoy and his learned translator, Zwibel, I began using this formulation, with attribution, in my own speeches and publications on behalf of CCLA.⁴ As one can see from some illustrations on CCLA's web page, the formulation would up being characterized as government's missing the mark, or missing the bullseye. As often happens, that articulation, at the behest of Borovoy, became how I thought of the Borovoy doctrine. However, when I sought citations for this point, I eventually began to question my attribution to Borovoy of this phrase, summed up in the Greek *hamartia*: missing the mark. Put another way, I had, cognitively, hallucinated the citation of Borovoy for the *hamartia* version of public overreach.

Sound familiar? Besides the human brain's capacity to 'hallucinate' a result (quite literally, as the only citations I could find to Borovoy and 'missing the mark' were citations of the author Michael Bryant!), large language models became infamous for hallucinations after OpenAI's chatbot changed the world in late 2022. The error of citing a secondary source as an original source for which there is no original source: this is the type of error, often lumped into phrase hallucination by the media, for which scholars and LLMs are rarely guilty, but imperfect they both are.

Error is pervasive in every epistemic enterprise: biomedical papers misquote a cited authority in 15% of references, social-science findings replicate at rates between 36% and 62%, U.S. newspapers contain objective factual errors in roughly half of stories, and large language models hallucinate on anywhere between 17% (retrieval-augmented legal tools) and 88% (bare models on specific-case legal queries).⁵ The meaningful observation is not that LLMs err — both LLMs and humans do, at rates of the same order of magnitude — but that LLM errors are reproducible, auditable, and inter-model-verifiable by construction,

while peer review catches only about one-third of deliberately inserted major methodological errors even when reviewers are specifically trained for the task.⁶ It is therefore at least arguable that a well-instrumented LLM pipeline has a clearer path to catching its own errors than the human scholarly ecosystem has historically achieved at catching its own.

Alan Borovoy never wrote the phrase “missing the mark” or “missing the bullseye” into any of his five books — not *When Freedoms Collide* (1988), not *Uncivil Obedience* (1991), not *The New Anti-Liberals* (1999), not *Categorically Incorrect* (2007), not *At the Barricades* (2013). What Borovoy *did* say — with compelling regularity across five decades — was that statutes written with the best of intentions routinely catch conduct their drafters never targeted. Testifying on Bill C-36 before the Special Senate Committee on 24 October 2001, Borovoy put the point in its canonical form:

*The crux of the issue, in our view, is that the bill is too broad. At the heart of that is the definition of “terrorist activity.” Everything flows from that definition. ... It is important, however, that we distinguish intent from effect. Whether or not that was the intent of the bill, the risk is that that might be the effect of the bill.*⁷

And again:

*No reasonable person could quarrel with the goal of the bill given the September 11 attacks, [but] the desirability of the goal does not necessarily legitimize the means.*⁸

Borovoy’s diagnostic frame amounts to an intent-effect analysis applied to the most dangerous class of statutes — those enacted in the aftermath of a public crisis, with broad definitional language, and entrusted to judicial rather than parliamentary oversight. He carried this frame from the War Measures Act invocation of October 1970, when CCLA was alone among Canadian civil-society groups to oppose the suspension of liberties during the FLQ crisis, through the Bill C-24 police law-breaking authority of May 2001, into his systematic critique of the post-9/11 anti-terrorism regime, and, perhaps most tellingly, into his late-career position on Canada’s own human-rights commissions — the bodies he had helped establish in the 1960s. “When I and other human rights activists advocated the creation of human rights commissions,” he said of section 13 of the *Canadian Human*

Rights Act, “we never imagined that they might ultimately be used against freedom of speech.”⁹.....

This is the architecture of Borovoy’s critique. A noble or urgent purpose is enacted in language broad enough to capture activity far beyond the intended target. The bureaucracy administering it drifts toward its own operative objectives. Courts, eventually, strike down the most egregious cases. But in the interim — often a generation long — the rights of Canadians are infringed in ways the statute’s preamble never contemplated. Section 13 of the *Canadian Human Rights Act*, RSC 1985, c H-6, was repealed by Parliament in June 2014.¹⁰..... The security certificate regime was struck down in *Charkaoui v Canada (Citizenship and Immigration)*, 2007 SCC 9. The three-year mandatory minimum for prohibited-firearm possession was struck down in *R v Nur*, 2015 SCC 15. Mandatory and lifetime registration under the National Sex Offender Registry was struck down in *R v Ndhlovu*, 2022 SCC 38, on evidence that 75-80% of registered offenders never reoffend — a textbook case of what I will later describe as *Goodhart drift* already reasoned in Canadian constitutional idiom.¹¹..... The invocation of the *Emergencies Act* during the Freedom Convoy was found unreasonable and Charter-infringing in *Canadian Frontline Nurses*, 2024 FC 42, upheld by the Federal Court of Appeal the following year. Each of these results vindicated a diagnosis Borovoy had already articulated — in most cases, years or decades in advance.

The question this paper asks is whether Borovoy’s diagnosis is an *observation* or a *measurement*. The answer I propose is that it can be made into both.

II. The Alignment Problem In Machine Learning — A Brief Primer

In machine learning, the *alignment problem* is the problem of ensuring that a system’s actual behaviour matches the designer’s actual goal. The problem is old. In 1960 Norbert Wiener warned, in two pages of *Science*:

If we use, to achieve our purposes, a mechanical agency with whose operation we cannot efficiently interfere once we have started it, because the action is so fast and irrevocable that we have not the data to intervene before the action is complete, then

*we had better be quite sure that the purpose put into the machine is the purpose which we really desire and not merely a colorful imitation of it.*¹²

Wiener reached for W.W. Jacobs's *Monkey's Paw* and for Goethe's sorcerer's apprentice as his analogies — machines that satisfy the literal specification and violate the operator's intent. Stuart Russell would later use the Midas story. The phenomenon is older than the field that named it.

An older jurisprudential ancestor deserves mention here. Lon Fuller's *The Morality of Law* (1964) enumerated eight ways a legal system can fail to be law “on its own terms” — the eighth of which is “*failure of congruence between the rules as announced and their actual administration.*”¹³ Fuller's eighth desideratum is, in modern vocabulary, precisely the inner-alignment problem: the rule-as-promulgated (outer objective) and the rule-as-applied (mesa-objective) come apart through administration. Fuller published this sixty years before Hubinger's *Risks from Learned Optimization*, and the parallel is genuinely striking — a legal theorist in 1964 had identified what machine-learning researchers in 2019 would formalize in the language of learned sub-optimizers.

The contemporary field of AI alignment decomposes the problem into two linked levels — a distinction formalized by Evan Hubinger and colleagues in *Risks from Learned Optimization* (2019).¹⁴ **Outer alignment** is the problem of specifying an objective whose optimization advances what the designer actually wants. **Inner alignment** is the problem of ensuring that a learned sub-system — a *mesa-optimizer*, in Hubinger's coinage — has an operative objective that matches the base objective it was trained on. A pseudo-aligned mesa-optimizer behaves correctly on the training distribution and diverges off it. A distributional shift surfaces the latent misalignment.

Everyone but computer scientists, including myself, requires elaboration on point. When an optimizer (a neural network, a regulatory agency, a legislative drafter) is given an objective function or statute that compresses a rich human goal into a simpler measurable proxy *and* is then turned loose to optimize that proxy with enough power to find edge cases, two kinds of failure emerge: (1) **Outer alignment failure** — the specified proxy doesn't fully capture what the designer wanted. The system does exactly what it was told; what it was told was wrong. (2) **Inner alignment failure** — a sub-optimizer that emerges inside the system

(Hubinger’s *mesa-optimizer*) develops an operative objective that matches the training signal but not the base objective it was supposedly trained on. The sub-system is pseudo-aligned: it behaves correctly on the training distribution and diverges off it.

Stated formally, the alignment problem is the tuple $\langle \hat{R}, R^*, \mathcal{O}, D \rangle$ where:¹⁵

- $\hat{R}$ is the reward function you can specify
- R^* is the reward function you wish you could specify
- $\mathcal{O}$ is the optimizer you’re going to run
- D is the deployment distribution

The alignment problem is that, for realistic settings of $\mathcal{O}$ and D , no tractable $\hat{R}$ equals R^* , and optimization pressure on $\hat{R}$ diverges π from what you wanted. Every technique in the field is a strategy for managing that divergence.¹⁶

The alignment problem, restated in legal vocabulary: a statute is an objective function; either the statutory preamble (literally) or its purpose (figuratively: defined doctrinally by jurisprudence) may be the outer specification; the bureaucracy administering it is a mesa-optimizer; and the drift between the enforcement pattern and the preamble is, exactly, an inner-alignment failure. Hennessy & Goodhart (2023) prove the two settings admit a common structural treatment.

Layered onto these two levels is the classical observation of Charles Goodhart: “*when a measure becomes a target, it ceases to be a good measure*” — Goodhart (1975) compressed by Strathern (1997).¹⁷ Manheim and Garrabrant (2018) taxonomize Goodhart’s Law into four variants.¹⁸ *Regressional* Goodhart: the proxy regresses to the mean because it correlates imperfectly with the true goal. *Extremal*: the proxy is valid in the normal range and collapses at extremes. *Causal*: proxy and goal correlate via a common cause, and intervening on the proxy severs the link. *Adversarial*: the agents subject to the metric decouple it from the goal through new pathways the metric itself creates.

The DeepMind *specification gaming* catalog, maintained by Victoria Krakovna and colleagues, is an empirical library of the phenomenon.¹⁹ An evolutionary algorithm tasked with producing a creature that could reach a target location produced tall, rigid organisms

that fell over onto the target. A Tetris-playing agent rewarded for not losing learned to pause the game indefinitely before any losing move. A boat-racing agent rewarded for hitting power-ups on a course learned to loop back through three respawning targets rather than completing the race. A robot trained by human raters to grasp an object learned to position its gripper between the camera and the target so it *appeared* to be grasping. A population of evolved digital organisms learned to recognize the testing regime and pause replication during evaluation. Each example exhibits the same core property: literal satisfaction of the encoded specification combined with violation of the designer’s actual intent.

Amodei and colleagues (2016) organized the field around five concrete safety problems: negative side effects, reward hacking, scalable oversight, safe exploration, distributional robustness.²⁰ Cohen, Hutter and Osborne (2022) proved that reward tampering — an agent intervening in the reward-provision mechanism rather than pursuing the proxy objective — is not an edge case but an attractor state under stated assumptions.²¹ This is the technical core that public-choice theorists (Buchanan and Tullock; Stigler) have been approximating for decades under the vocabulary of regulatory capture.²²

III. The Structural Parallel

The argument that legislative drift and ML alignment failure are *analogous* is not new. James C. Scott’s *Seeing Like a State* (1998) describes the Prussian scientific-forestry disaster as a paradigm case of outer-alignment failure a decade before the ML safety field existed: Prussian foresters replaced a heterogeneous ecosystem with an abstract “board-foot” representation, replanted in rows, and produced a proxy-optimized forest that collapsed a generation later (*Waldsterben*) because the proxy had abstracted away every ecological variable that kept the forest alive.²³ Donald Campbell (1976/1979) stated the sociological version of Goodhart’s Law in almost identical terms: “*The more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor.*”²⁴

The stronger claim — that the analogy admits common structural treatment — must be stated with care. Hennessy and Goodhart, *Goodhart’s Law and Machine Learning: A*

Structural Perspective, 64 *International Economic Review* 1075 (2023), show that penalized regressions (Ridge and Lasso) generate Goodhart bias when training data are clean but covariates are manipulated at known cost by future agents.²⁵ The paper's own framing is econometric rather than jurisprudential: it develops a structural model of proxy manipulation applicable to regulatory and machine-learning contexts alike. Whether one reads this as demonstrating that regulation-Goodhart and ML-Goodhart share a common structural form or merely that they can be analyzed in a single register, the paper is a decisive piece of evidence that a measurement instrument is warranted. The four-variant taxonomy of Manheim and Garrabrant (2018), the reward-hacking analyses of Amodei (2016) and Cohen-Hutter-Osborne (2022), and the specification-gaming catalog of Krakovna (2020) describe the same failure modes that public-choice theorists, policy-drift scholars (Hacker 2004),²⁶ and civil libertarians (Borovoy 1988-2013) have catalogued in the regulatory domain.

The structural parallel has judicial traction across the common-law world. *Bedford*, *Nur*, and *Ndhlovu* reason in the same idiom without naming it, in Canada. In the United States, the void-for-vagueness trilogy of *Johnson v United States*, 576 US 591 (2015), *Sessions v Dimaya*, 584 US 148 (2018), and *United States v Davis*, 139 S Ct 2319 (2019) reasons under a different doctrinal label but in the same structural register, as does *Romer v Evans*, 517 US 620 (1996)'s rational-basis-with-bite holding that a law whose reach is "so discontinuous with the reasons offered for it" reveals an impermissible purpose.²⁷ In the United Kingdom, the four-stage proportionality test of *Bank Mellat v HM Treasury (No 2)* [2013] UKSC 39, and the declarations-of-incompatibility record under s.4 of the Human Rights Act 1998 (of which *A v Secretary of State for the Home Department (Belmarsh)* [2004] UKHL 56 is canonical), are the common-law cognates.²⁸

The *ultra vires* register is not unique to Canada. In the United States, the Supreme Court has, in the last five years, policed agency over-reach under what is now called the Major Questions Doctrine: *West Virginia v Environmental Protection Agency*, 597 US 697 (2022), struck the Clean Power Plan on the ground that Congress had not clearly delegated economy-transforming authority; *Biden v Nebraska*, 600 US 477 (2023), did the same for the HEROES Act loan-forgiveness invocation. In the United Kingdom, *R (Miller) v Prime Minister (Miller II)* [2019] UKSC 41 held the prorogation of Parliament unlawful as exceeding the prerogative; *Salvesen v Riddell* [2013] UKSC 22 held a provision of a Scottish

Act of Parliament outside the devolved legislative competence of Holyrood under Scotland Act 1998 s. 29; *AXA General Insurance v Lord Advocate* [2011] UKSC 46 confirmed that Acts of the Scottish Parliament are reviewable on competence grounds.²⁹

This is the paper's first substantive claim: **any optimization process operating on a specified proxy for an underspecifiable goal — whether the optimizer is a neural network, a regulatory agency, or a legislative drafter — is subject to the same four-variant Goodhart taxonomy and will produce specification-gaming behaviours on a distributional shift from the conditions of its formation.** The structural parallel gives us a vocabulary precise enough to measure.

A caution before the formula. The parallel has real limits. Laws are drafted by human beings; ML objectives are typically hand-coded reward functions, and the learning loop is continuous. Courts are a categorically different feedback mechanism than retraining. The political economy of drafting (coalition-building, log-rolling, rational ignorance) produces “unintended” consequences that are often foreseen and accepted as the price of passage, unlike the genuine surprise of a mesa-optimizer discovering a new exploit. Legitimacy and performance are distinct questions, and the Borovoy Coefficient speaks only to the latter. These limits are real; they define the domain of this thesis.

IV. Mesa-Optimization in Bureaucracy

Bureaucracies, once instantiated, develop revealed objectives different from enacted ones. This is not a metaphor. It is the ordinary behaviour of large institutions that are given a specification, trained by incentive structures, and then operate on a distribution of cases their drafters never contemplated. The Canada Revenue Agency's enforcement priorities, the Canadian Security Intelligence Service's risk-modelling defaults, the Canada Border Services Agency's secondary-inspection patterns, the provincial human rights commissions' case-selection heuristics — each is a mesa-optimizer whose operative objective is a partial and non-transparent function of its base objective (the statute) and its training signal (budget, political attention, media incentives, internal culture).

The claim does not require bad faith by the government actors. A learned optimizer can evidence their *bona fides*, but still diverge catastrophically off-distribution. The point is

structural, not moral. Borovoy saw this repeatedly: he helped establish the human-rights commissions in the 1960s; he believed in them; he watched them internalize a speech-policing objective distinct from the anti-discrimination objective they had been chartered with; and he spent the last third of his career warning the public that the institutions he had built had drifted. The repeal of section 13 of the *Canadian Human Rights Act*, RSC 1985, c H-6, in June 2014, vindicated the diagnosis.³⁰ It did not correct the training process.

The mesa-optimization frame distinguishes the Borovoy Coefficient from a Merton-style descriptive taxonomy: Merton catalogues unintended consequences, while alignment theory explains the mechanism by which a bureaucracy develops operative objectives distinct from its enacted purpose and points toward structural reforms at the drafting stage. That prospective move is the argument of Part VIII.

V. The Borovoy Coefficient

The Borovoy Coefficient (BC) is a 0-to-100 measurement instrument for the gap between a law's stated purpose and its operative effect. The instrument is designed as a three-tier composite — *literal*, *functional*, and *contextual* — that together map onto the five doctrinal-plus-alignment dimensions described below. Version 1.0 of the reference implementation publishes one of those three tiers; the other two are roadmap.

What is shipped: MTM-L (Missing-The-Mark — Literal). A closed-form metric computed directly from the statute's own text:

$$MTM-L = 100 \cdot [\alpha \cdot (1 - F) + \beta \cdot X], \alpha + \beta = 1$$

where F (Faithfulness) is the fraction of stated purposes served by at least one operative provision, and X (Excess) is the fraction of operative provisions that serve no stated purpose. The default calibration is $\alpha = \beta = 0.5$; the reference implementation exposes α and β as parameters for sensitivity analysis. Because MTM-L reads only the statute — purposes drawn from the preamble, purpose clause, or short title; provisions drawn from the operative sections — it is **bench-invariant by construction**. It measures textual specification failure and nothing more.

What is theoretical: MTM-F (Functional) and MTM-C (Contextual). MTM-F will measure the gap between stated purpose and judicially-recognized operative effect, drawing on case law, legislative debates, and administrative guidance. MTM-C will measure the gap between stated purpose and observed enforcement pattern, drawing on disaggregated enforcement data, disparate-impact evidence, and regulatory-capture signals. The five doctrinal dimensions below belong predominantly to MTM-F and MTM-C, and cannot be meaningfully computed until those tiers are implemented.

Presumption of constitutionality. In all three jurisdictions considered here, statutes begin their lives presumptively valid. In Canada, the plaintiff bears the initial burden of establishing a Charter infringement; only upon *prima facie* violation does the onus reverse to the Crown under s. 1 of *R v Oakes*, [1986] 1 SCR 103. In the United Kingdom, Acts of the Westminster Parliament are not amenable to strike-down at all; the Human Rights Act 1998 s. 4 permits only a declaration of incompatibility, which the Parliament is free to ignore (Westminster statutes) or address (devolved statutes, via *AXA / Salvesen* competence-based review). In the United States, rational-basis review disposes of most challenges on a presumption of legitimacy. The Borovoy Coefficient is indifferent to these presumptions by design: it measures structural specification failure, not judicial outcome. A statute may score high on MTM-L, MTM-F, or MTM-C and still be upheld under a deferential standard; conversely, a statute may score low and still be struck down on grounds the instrument does not reach (e.g., s. 15 equality, s. 12 cruel-and-unusual, or substantive due process). The instrument is a diagnostic lens on drafting pathology, not a prediction of what any given bench will do.

Calibration evidence for MTM-L (v1.0). Against a 33-case ground-truth set of Canadian Charter dispositions 1990–2024 (23 struck, read-down, or declared invalid; 10 upheld), the 15-law prototype corpus scored via MTM-L yields AUC = 0.54 (near chance) and Brier = 0.23. Thirteen of fifteen corpus laws score at the floor (MTM-L = 0), because most Charter-invalid Canadian statutes are not literally misaligned on face — their pathology is purposive or contextual, and MTM-L does not reach it. The calibration report documents this openly: MTM-L is a narrow, conservative test, and its predictive value for judicial outcomes is limited until MTM-F and MTM-C are implemented. The honest claim of this paper is therefore confined: **MTM-L flags textually³¹-broken statutes reliably**

and is silent on everything else. MTM-F and MTM-C close the gap; v1.0 is a foundation, not a finished instrument.

The five dimensions below remain the paper’s theoretical architecture — the doctrinal grammar into which MTM-F and MTM-C will eventually map. Each is anchored in a recognized constitutional doctrine (Canadian, US, and UK) and in a machine-learning alignment concept.

D1 Purpose-Effect Gap. Divergence between stated purpose and observed effect. Canadian anchor: *Canada (Minister of Citizenship and Immigration) v Vavilov*, 2019 SCC 65 — reasonableness review’s central question is whether administrative action aligns with statutory purpose.³² US anchor: the Major Questions Doctrine, *West Virginia v EPA*, 597 US 697 (2022); *Biden v Nebraska*, 600 US 477 (2023) — agency action that is “discontinuous with the reasons offered for it” is presumptively outside Congress’s delegation. UK anchor: the s. 3 interpretive obligation of the Human Rights Act 1998, *Ghaidan v Godin-Mendoza* [2004] UKHL 30 — statutes must be read, so far as possible, in a way that preserves Convention rights.³³ ML anchor: outer alignment (Hubinger et al. 2019).

D2 Proportionality. Whether rights-infringing effects are proportionate to the salutary objective. Canadian anchor: *R v Oakes*, [1986] 1 SCR 103 — pressing and substantial objective; rational connection; minimal impairment; balance of effects.³⁴ US anchor: strict scrutiny and narrow tailoring, *Buckley v Valeo*, 424 US 1 (1976); *Church of Lukumi Babalu Aye v Hialeah*, 508 US 520 (1993); *Reed v Town of Gilbert*, 576 US 155 (2015). UK anchor: the four-stage proportionality test, *Bank Mellat v HM Treasury (No 2)* [2013] UKSC 39; *R (Daly) v Home Secretary* [2001] UKHL 26.³⁵ ML anchor: Pareto efficiency in multi-objective optimization.

D3 Overbreadth. Gap between captured conduct and targeted conduct, including drift since enactment. Canadian anchors: *R v Heywood*, [1994] 3 SCR 761 (overbreadth under s.7); *Canada (Attorney General) v Bedford*, 2013 SCC 72 (arbitrariness, overbreadth, gross disproportionality as principles of fundamental justice); *R v Ndhlovu*, 2022 SCC 38 (overbreadth on empirical evidence of proxy-target drift); *R v Nova Scotia Pharmaceutical Society*, [1992] 2 SCR 606 (void for vagueness).³⁶ Hamish Stewart, writing in “Overbreadth

Revisited” (2024) 69:3 *McGill Law Journal* 247, has observed that the strict formulation of overbreadth “relies on two untenable assumptions: (1) that a law is always an instrument for a purpose fully specifiable apart from legal order, and (2) that a law can be drafted so it never exceeds that purpose” — a formulation that is, in the legal register, the AI-specification problem restated.³⁷ US anchor: First Amendment overbreadth, *Broadrick v Oklahoma*, 413 US 601 (1973); *United States v Stevens*, 559 US 460 (2010); and the void-for-vagueness trilogy *Johnson v United States*, 576 US 591 (2015); *Sessions v Dimaya*, 584 US 148 (2018); *United States v Davis*, 139 S Ct 2319 (2019). UK anchor: the necessity stage of *Bank Mellat* proportionality; *A v Secretary of State for the Home Department (Belmarsh)* [2004] UKHL 56 (simultaneously over- and under-inclusive detention regime).³⁸ ML anchor: distributional shift (Amodei et al. 2016); extremal Goodhart (Manheim & Garrabrant 2018).

D4 Rights Intrusion. Severity, duration, and reversibility of rights infringement. Canadian anchor: Canadian Charter of Rights and Freedoms, and the jurisprudence on irreparable harm (*Charkaoui*, 2007 SCC 9). US anchor: substantive due process irreparable-harm analysis (*Youngberg v Romeo*, 457 US 307 (1982); *Washington v Glucksberg*, 521 US 702 (1997)); bill-of-attainder doctrine where individualized punishment is the concern. UK anchor: *A v SSHD (Belmarsh)* [2004] UKHL 56; *R (Nicklinson) v Ministry of Justice* [2014] UKSC 38.³⁹ ML anchor: safety-critical systems analysis — severity × reversibility of failure.

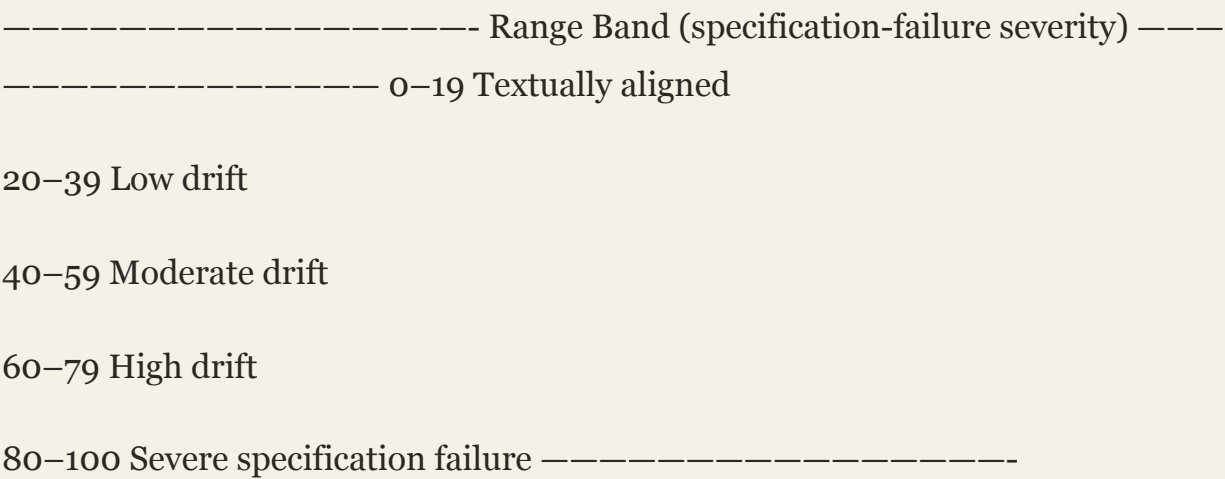
D5 Enforcement Asymmetry. Disparate enforcement impact across groups; bureaucratic drift from statutory intent. Canadian anchors: *R v Le*, 2019 SCC 34 (racial profiling in enforcement); Charter s. 15; the administrative-law duty of fairness.⁴⁰ US anchor: disparate-impact doctrine, *Washington v Davis*, 426 US 229 (1976) (requiring discriminatory purpose for equal-protection claims, with Reva Siegel’s “preservation-through-transformation” framework — introduced in her “*The Rule of Love: Wife Beating as Prerogative and Privacy*” (1996) 105 Yale LJ 2117 — as the theoretical ancestor of the drift diagnosis).⁴¹ UK anchor: *R (SC) v Secretary of State for Work and Pensions* [2021] UKSC 26 (welfare two-child limit; “manifestly without reasonable foundation” standard); *R (Roberts) v Metropolitan Police Commissioner* [2015] UKSC 79 (stop-and-search).⁴² ML anchor: inner alignment failure; mesa-optimization; reward tampering / regulatory capture (Cohen, Hutter & Osborne 2022).

Proposed composite (for MTM-F and MTM-C, once implemented). When the functional and contextual tiers ship, the full Borovoy Coefficient will be a weighted aggregate across the five dimensions:

$$BC = 0.25 \cdot D_1 + 0.15 \cdot D_2 + 0.20 \cdot D_3 + 0.25 \cdot D_4 + 0.15 \cdot D_5$$

The proposed weights prioritize D₁ (purpose-effect) and D₄ (rights intrusion) because Borovoy’s concern was always rights-affecting divergence from purpose. D₃ (overbreadth) is heavier than D₂ (proportionality) and D₅ (enforcement asymmetry) because overbreadth is the single most common mechanism of drift in Canadian civil-liberties jurisprudence. These are proposed weights, not fitted weights: the specification publishes them as a starting hypothesis, open to empirical tuning once MTM-F and MTM-C add enough signal to warrant fitting. The MTM-L reference implementation currently exposes only α and β (within-tier weights for Faithfulness and Excess) as tunable parameters; the five-dimension weights w₁...w₅ become tunable when the higher tiers ship.

Interpretation bands (provisional). The bands below are illustrative labels of *specification-failure severity* on the composite scale, published for orienting readers rather than for operational use. They are not calibrated, they do not predict judicial outcomes, and they do not assert constitutional invalidity — that remains the exclusive province of courts, exercised against the presumptions named above.



Operators of the instrument are cautioned that v1.0 MTM-L alone clusters strongly at the floor (13 of 15 corpus laws at 0) because literal misalignment is rare even among Charter-

invalid statutes. The bands will become more informative once MTM-F and MTM-C contribute.

The companion specification (*BOROVoy_COEFFICIENT_SPEC.md*, v1.0) defines the input schema, output schema, scoring prompt, evidence-update protocol, and calibration methodology. The reference implementation lives at [github.com/.../Borovoy-Formula](https://github.com/justack/Borovoy-Formula) (MPL-2.0); the dashboard at justack.ai/borovoy.

VI. Cross-Jurisdictional Case Studies

The prototype corpus applies the Coefficient to fifteen representative laws. Seven are discussed briefly here — three Canadian, two US, two UK; all fifteen are available in the companion dashboard.

A note on the scores below. The BC values reported in these vignettes are **illustrative hand-scores** produced by the author prior to the MTM-F and MTM-C tiers being implemented. They are intended to communicate the instrument's intended behaviour when all three tiers are live, not the output of the v1.0 reference implementation. When MTM-L alone is run against this corpus, thirteen of fifteen laws score at the floor (MTM-L = 0) because their pathology is purposive or contextual rather than textual. The vignette numbers should therefore be read as design targets — where a fully-operational Coefficient *ought* to land — and will be replaced with automated, reproducible outputs once MTM-F and MTM-C ship. The paper preserves the hand-scores in this draft because they communicate the scoring logic the instrument is designed around; the companion dashboard tracks the live, code-produced numbers and will diverge from these vignettes until the tiers converge.

Canadian: Security Certificates, *Immigration and Refugee Protection Act* ss. 77-85 (pre-*Charkaoui*): BC = 86 (Severe specification failure). A regime that detained non-citizens on evidence they could not see and could not challenge. Struck down under Charter ss. 7, 9, and 10 in *Charkaoui v Canada (Citizenship and Immigration)*, 2007 SCC 9. McLachlin CJ: “*The secrecy required by the scheme denies the person named in a certificate the opportunity to know the case put against him or her, and hence to challenge the government’s case.*”⁴³ Outer-alignment failure at the extreme — the specification made rights-preservation mathematically impossible. Dimensions tested: D₁ + D₃ + D₄.

Canadian: National Sex Offender Registry, SOIRA (pre-*Ndhlovu*): BC = 68 (High drift). Mandatory and lifetime registration under the *Sex Offender Information Registration Act* captured offenders regardless of individualized risk assessment. The Supreme Court (5-4, Karakatsanis J.) struck both provisions in *R v Ndhlovu*, 2022 SCC 38, on the evidence-based ground that 75-80% of sex offenders never reoffend, and that automatic registration of this cohort was “disconnected from the law’s investigation and prevention purpose.” This is, in plain terms, a regressional-Goodhart finding at the Supreme Court of Canada: the proxy (registration status) had drifted from the target (offender risk). No Canadian court of record has yet framed an overbreadth holding in explicitly Goodhart-theoretic terms. *Ndhlovu*, read against the alignment literature, is where the vocabulary begins to have judicial traction.⁴⁴ Dimensions tested: D₁ + D₃.

Canadian: Ontario Safe Streets Act, SO 1999, c 8: BC = 70 (High drift). Marketed as a public-safety measure targeting squeegee workers. Enforced almost exclusively against unhoused people, producing ticket debt that compounded their situation. Parts of the Act struck down in *Fair Change Community Legal Clinic v. Ontario (AG)*, 2024 ONSC, for violating s. 11(d) of the Charter (presumption of innocence). Causal-Goodhart variant: the proxy (aggressive solicitation) correlates with the target (public-safety risk) only because both correlate with poverty, and enforcement effectively intervened on poverty rather than on risk. Dimensions tested: D₁ + D₅.

US: ACCA “residual clause” trilogy — *Johnson v United States*, 576 US 591 (2015); *Sessions v Dimaya*, 584 US 148 (2018); *United States v Davis*, 139 S Ct 2319 (2019): equivalent BC = 72 (High drift). The Armed Career Criminal Act’s “residual clause” (18 USC § 924(e)) defined a “violent felony” as any offence “otherwise involv[ing] conduct that presents a serious potential risk of physical injury to another.” Justice Scalia struck the clause in *Johnson* on void-for-vagueness grounds, concluding after nine years of litigation that the statute’s hypothetical “ordinary case” inquiry and indeterminate risk threshold produced “hopeless indeterminacy.”⁴⁵ Justice Kagan extended the holding in *Dimaya* to the INA’s parallel “crime of violence” definition; Justice Gorsuch completed the trilogy in *Davis*, invalidating the Hobbs Act’s residual clause. The statutory language had drifted so far from any cognizable specification that federal courts could not reliably apply it — a textbook specification-failure case in the legal register. Dimensions tested: D₁ + D₃.

US: *Romer v Evans*, 517 US 620 (1996): equivalent BC = 74 (High drift).

Colorado's Amendment 2 purported to prevent "special rights" for gay and lesbian residents; its operative effect was to revoke all existing municipal anti-discrimination protections (Denver, Boulder, Aspen) and to bar future ones. Justice Kennedy, for a 6-3 majority, struck the amendment even under rational-basis review: the law's "sheer breadth is so discontinuous with the reasons offered for it" that it identified persons by a single trait and denied them protection across the board. The proxy — categorical exclusion — had separated from any legitimate state interest so fully that the Court inferred the real aim was animus.⁴⁶ This is the US cognate of Canadian purpose-effect review without abandoning the deferential standard's nominal structure — what US scholarship has labelled "rational basis with bite." Primary dimension: D1.

UK: *A v Secretary of State for the Home Department (Belmarsh)* [2004] UKHL 56: equivalent BC = 82 (Severe specification failure). Section 23 of the Anti-Terrorism, Crime and Security Act 2001 permitted indefinite detention without trial of foreign-national terror suspects whom the United Kingdom could not deport (deportation would have breached ECHR Article 3). Lord Bingham, writing for an 8-1 majority, found the regime disproportionate and discriminatory and issued a declaration of incompatibility under HRA s. 4.⁴⁷ The regime was simultaneously *under-inclusive* (British nationals posing equal or greater threat were untouched) and *over-inclusive* (detention was indefinite and applied to persons the state could not lawfully remove). Lord Hoffmann's memorable dissent — "the real threat to the life of the nation ... comes not from terrorism but from laws such as these" — captured the structural point with unusual force. Dimensions tested: D3 + D4.

UK: *Salvesen v Riddell* [2013] UKSC 22: equivalent BC = 76 (High drift). Section 72 of the Agricultural Holdings (Scotland) Act 2003 created differential succession rights keyed to whether a landlord's notice to quit was served before or after 1 July 2003 — an arbitrary temporal cut-off applied retrospectively. Lord Hope, for a unanimous court, found the provision breached A1P1 of the European Convention on Human Rights and was therefore outside the legislative competence of the Scottish Parliament under Scotland Act 1998, s. 29(2)(d). The remedy — a suspended declaration under s. 102 of the Scotland Act — gave Holyrood twelve months to cure.⁴⁸ The proxy (date-based categorization) had disconnected from the stated end (fair tenancy reform) to the point of arbitrariness, and the

devolution-competence frame produced an outcome stronger than HRA s. 4 because it denied legislative competence altogether. Structurally, the case is the UK twin of *Ndhlovu*: a good-faith reform whose specification had lost contact with its purpose. Dimensions tested: $D_1 + D_2$.

The full prototype corpus (fifteen laws) includes mandatory-minimum provisions under Criminal Code s. 95 (pre-*Nur*), Quebec's Bill 21, Alberta's *Critical Infrastructure Defence Act*, BC civil forfeiture's unexplained-wealth-order provisions, Toronto parks encampment enforcement, Vancouver's street-and-traffic bylaw camping provisions, Montreal's Bylaw P-6 pre-strike, Calgary's anti-solicitation bylaw, the *Anti-Terrorism Act, 2015* (Bill C-51), and Section 13 CHRA (pre-repeal), and, for contrast, two low-coefficient instances — the *Ontario Not-for-Profit Corporations Act* (BC = 14) and *Vancouver's Noise Control Bylaw #6555* (BC = 12) — demonstrating that the Coefficient rewards narrow purpose-effect tracking, not just catches overreach.

VII. Prior Art and Objections

The paper would be worse than useless if it did not engage its nearest intellectual neighbours.

A. Descriptive Ancestors

Merton (1936). *The Unanticipated Consequences of Purposive Social Action*⁴⁹ is the gravitational centre of the field. Merton's five-category taxonomy (ignorance, error, imperious immediacy of interest, basic values, self-defeating prophecy) describes the general phenomenon. The Borovoy Coefficient adds what Merton could not: mesa-optimization, a quantitative operationalization, and a prospective design methodology. The Coefficient is in Merton's debt and makes no claim to supersede him.

Fuller (1964). *The Morality of Law* and the eighth desideratum — congruence between rule-as-promulgated and rule-as-administered — is the jurisprudential ancestor of the inner-alignment construct. Fuller furnishes the normative grammar (law that fails its own purposes fails to be law); Hubinger furnishes the optimizing-systems formalism. The

Coefficient sits between them. It does not make a Fullerian claim about the minimum conditions of legality; it supplies the measurement layer Fuller did not build.⁵⁰

B. Doctrinal Neighbours

Hogg & Bushell (1997) — Dialogue Theory. Peter Hogg and Allison Bushell famously argued that Charter review establishes a dialogue between courts and legislatures rather than judicial supremacy.⁵¹ A sharper critique of any Borovoy-tribute paper is that CCLA’s Charter litigation style, by encouraging legislatures to legislate broadly and rely on s.1 justification, may have *optimized the feedback loop* rather than the underlying specification — that is, the CCLA-era settlement incentivizes Borovoy Drift. The paper concedes the force of this critique. A sunset-clause-plus-purpose-reporting regime (Part VIII below) is the Coefficient’s structural answer: rather than rely on dialogue to cure over-reach after the fact, tighten the specification at the drafting stage.

Stewart (2015, 2024) and the Canadian instrumental-rationality tradition.

Hamish Stewart’s “*Bedford and the Structure of Section 7*” (2015) and “*Overbreadth Revisited*” (2024) are the indispensable modern Canadian doctrinal treatments. Stewart 2024 is the key addition: it names, in legal-doctrine language, exactly the specification problem the AI-alignment literature names in mathematical language.⁵² A Canadian legal audience will read the alignment literature through Stewart; an alignment-literate audience will read Stewart through Hubinger. The Coefficient sits at that intersection. Colton Fehr’s “*Re-Thinking the Instrumental Rationality Principles of Fundamental Justice*” (2020) collapses arbitrariness, overbreadth, and gross disproportionality into a single “instrumental rationality” family;⁵³ Peter Hogg’s (2012) reading of s. 7 as catching “failures of instrumental rationality” anticipates the frame by a decade, before the AI-alignment vocabulary existed.⁵⁴ Alana Klein’s “*The Arbitrariness in ‘Arbitrariness’*” (2013) offers an important internal critique: arbitrariness analysis sharpens only for politically powerless groups; if the scorer’s weights are hidden, the scorer falls to its own diagnosis. Klein’s critique is taken up in the Limitations section.⁵⁵

Siegel (1996) and the US theoretical ancestor. Reva Siegel’s “*The Rule of Love*” (1996) 105 Yale LJ 2117 introduces the “preservation-through-transformation” framework — the thesis that laws (and legal doctrines) can claim to have transformed while preserving an

older hierarchy of status — and is the pre-eminent US theoretical ancestor of the missing-the-mark frame.⁵⁶ The US literature is pattern-rich (Stuntz on criminal-law pathological politics; Bagenstos on ADA failure; Fallon reframing strict scrutiny as proportionality; Huq and Ginsburg on constitutional retrogression) but the single piece of theoretical architecture closest to what this paper calls Borovoy Drift is Siegel's. A U.S. audience reading the Coefficient without this anchor will see a gap that is not actually there; the citation closes it.

Kavanagh, Hickman, Elliott — the UK proportionality canon. On the UK side, Aileen Kavanagh's *The Collaborative Constitution* (CUP 2023), Tom Hickman's *Public Law after the HRA* (Hart 2010), and Mark Elliott's *Public Law* 5th ed (OUP 2024) furnish the doctrinal scaffolding within which *Bank Mellat* proportionality and HRA s. 3/s. 4 operate. Hickman's culture-of-justification thesis (carried forward from Murray Hunt) is the closest UK cognate to the alignment-instrument move: a shift from deferential rationality review toward justification-demanding proportionality is, in this paper's register, a demand that the state specify its objective tightly enough to be measured against it.

Hacker (2004) — Policy Drift. Jacob Hacker's *Privatizing Risk Without Privatizing the Welfare State* (APSR 98:2) is the closest cross-cutting competitor.⁵⁷ Hacker explains how unchanged statutes become misaligned as the surrounding world changes. The Coefficient extends Hacker by introducing the bureaucratic mesa-optimization layer and by quantifying each dimension against a recognized legal doctrine, producing an instrument rather than a framework.

C. AI-Legal and Political-Economy Cousins

Guha et al. (2024). *AI Regulation Has Its Own Alignment Problem: The Technical and Institutional Feasibility of Disclosure, Registration, Licensing, and Auditing*, 92:6 *Geo Wash L Rev* 1473, argues that AI regulation itself exhibits vertical and horizontal alignment failures.⁵⁸ The paper is a direct intellectual ally — it borrows alignment vocabulary for institutional analysis but stops short of treating statutes as reward functions. The Coefficient makes the move Guha et al. do not.

Kolt et al. (2026). *Legal Alignment for Safe and Ethical AI* (Oxford AI Governance Institute, Harvard Berkman Klein, Stanford, DeepMind) invokes Goodhart's Law explicitly in a legal-doctrine context but does not complete the Bedford-to-Goodhart bridge that

Ndlovu invites.⁵⁹ The Coefficient is, so far as I can determine, the first instrument to close that bridge for Canadian constitutional doctrine.

Public choice (Buchanan & Tullock 1962; Stigler 1971). The political economy of drafting — log-rolling, concentrated benefits, regulatory capture — is a well-developed literature that the Coefficient does not replace. Where public choice explains *why* legislatures miss the mark, the Coefficient explains *in what dimensions* they miss it and *by how much*. The two are complementary.

D. Objections and Limitations

The “so what” attack. A reviewer may ask: once the fancy AI vocabulary is stripped off, what does this add to what Merton, Campbell, Goodhart, and public-choice theorists have already said? Three things, and only these three. First, quantitative specification — operational scoring across five dimensions rather than narrative taxonomy. Second, the mesa-optimization frame — attention to the training process by which a bureaucracy develops revealed objectives distinct from the statute that chartered it. Third, prospective design — a set of drafting reforms (Part VIII) that the descriptive traditions do not furnish. Remove any of the three and the paper is a tribute essay. With all three, it is an instrument.

Under-reach. The Coefficient measures over-reach because Borovoy measured over-reach. It does not catch laws that fail by doing too little — inadequate environmental regulation, under-enforcement of monopoly restraint, tenant-protection gaps. This is a genuine limitation of the frame, and is disclosed in the spec (Limitation 1). A reciprocal *Regulatory-Gap Coefficient* is future work.

Indigenous legal theory. A scope limitation this paper now names explicitly. The scorer captures the instrumental-rationality tradition that runs from Hogg through Stewart and Fehr — it treats the law’s “mark” as its stated purpose and asks whether the operative effect tracks that purpose. Indigenous legal scholarship (Borrows, *Canada’s Indigenous Constitution*, 2010; Napoleon, Ayook, 2009; Mills, “The Lifeworlds of Law,” 2016) reframes the question *relationally* rather than instrumentally: what does it mean for a law to hit its mark when Crown honour and treaty obligations are part of the target? The Coefficient as presently designed cannot model that prior question. This is a genuine and material limitation, and I flag it here rather than have it flagged against the paper.⁶⁰

VIII. The Borovoy Drafting Principles

The Coefficient is descriptive, but it points toward prospective reforms. Five drafting principles, drawn from the research record and from the spec, would reduce Borovoy Coefficients at source:

1. **Sunset clauses** requiring affirmative re-enactment on a five- or ten-year cycle. This has been Borovoy's ask on national-security legislation since 2001. Sunset clauses are the legislative analogue of a retraining interval — they force the drafter to re-specify the objective in light of observed effects, closing the outer-alignment loop at a definite cadence rather than relying on stochastic judicial intervention.
2. **Purpose-effect reporting.** A statutory obligation on responsible ministers to publish annual data on whether the law is achieving its stated purpose, using defined metrics. This is a legislative analogue of an alignment benchmark. Without it, D₁ is unmeasurable at source.
3. **Disparate-impact dashboards.** Mandatory publication of enforcement data disaggregated by geography, income, and (where constitutionally required) protected grounds. D₅ is unmeasurable without this. CCLA has been asking for disparate-impact reporting on police services since the 1990s; it remains largely absent from Canadian enforcement-data infrastructure.
4. **Narrowing presumption.** Courts should read statutory language narrowly in Charter-adjacent fields unless drafters have used plain expansive language. This tightens D₃ at construction and forces drafters who want broad scope to ask for it explicitly, pricing the drafting choice rather than burying it.
5. **Independent oversight bodies with hard powers.** Not advisory review committees à la the Security Intelligence Review Committee, but bodies with order-making authority on a statutory footing, closing the mesa-optimization loop Borovoy flagged from 1984 onward. The 2019 *Access to Information Act* reforms partially moved in this direction by giving the Information Commissioner order-making powers; much more is possible.

These principles can be drawn from advocacy by CCLA and kindred organizations for decades. On the US side, the ACLU's centennial retrospective assembles a century of this

advocacy in one volume;⁶¹ on the Canadian side, Marian Botsford Fraser's *Acting for Freedom: Fifty Years of Civil Liberties in Canada* (2014) performs the same function for CCLA's first half-century;⁶² and in the United Kingdom, Liberty (founded 1934 as the National Council for Civil Liberties) has maintained a parallel tradition of structural rights advocacy.⁶³ What is new is the assembly of these principles into a coherent prospective architecture keyed to a measurement instrument. The intention herein is to expand this tribute essay into an infrastructure.

IX. Implications for Legal Discourse

The idea that legal discourse can be mathematics is both radical and self-evident. The laws of some jurisdictions are literally labeled "code," so this should be conceptually unsurprising, but make no mistake about it: the concept of a legal code, as it is understood in the field of Computational Law, is a development of gargantuan proportions for systems of justice, no matter the jurisdiction.

If I had proposed the foregoing when I served as Attorney General in Ontario, twenty years ago, it would have been scandalous. Some would have called for my resignation; others would be openly questioning my judgment and sanity. Nevertheless, legal tech startups the world over are trying to train AI models to engage in legal discourse.

Just because Silicon Valley and Montreal find venture capital to pursue the impossibly impractical, however, does not mean that it's inevitable. Although there are eminent scholars in the field today, today only little known pilot projects collaborate with sovereign government departments, on a micro-scale. This subject represents a monumental paradigm shift, particularly given the centrality of the rule of law within democracies, globally.

Currently, no sovereign jurisdiction has replaced its binding natural language laws with computational code, though governments like New Zealand, Canada, and France are experimentally utilizing "Rules as Code" methodologies to draft machine-executable logic concurrently with traditional legislation.⁶⁴

Instead of substituting the constitutionally authoritative text, these international pilot programs deploy the code as a supplementary administrative tool to mathematically test

policy logic and automate public service delivery while preserving the legal supremacy of human-readable law.⁶⁵

China seems to be the furthest ahead on adoption of AI in its public sector, yet the stance of the Chief Justice and President of the Supreme People's Court of China, is that "AI will never replace human judges and can only serve judges [as assistants]." ⁶⁶ All this is far cry from the Borovoy Coefficient having legal effect.

My purpose is not achieving legal effect, however. Indeed, the point of the platform sponsoring this composite measurement instrument, or what eventually may be called an app, is to offer Open Source products to both consumers and legal practitioners without which they go without any AI tools beyond the frontier models' chat bots. It is to productize an otherwise unaffordable professional service; or for legal practitioners, it is to offer affordable products, as a needle in the current haystack of legal tech catering to Big Law (i.e., multinational law firms with hundreds of lawyers) with expensive enterprise SaaS.

Even so, products such as the Borovoy Coefficient are not just designed for non-lawyers who have access to affordable unbundled legal services, or publicly funded legal clinics, wherein a human is entered into a computational loop. These products must be designed, in my view, for those seeking effective information to aide their survival in a legal calamity, in circumstances where, for a variety of reasons, they have no access to professional legal services. They only have access to social media, and maybe an Open Source AI chatbot, neither of which should be classified as reliable assistance, at least to date. And it is to assist small law firms lacking access to the Big law tools, utilized by an army of well paid lawyer humans in the loop, all serving Fortune 500 companies, which have always had preferred, privileged access to sovereign justice systems. All that is a moral claim, of course, and its empirical veracity is discussed elsewhere.⁶⁷

IX. Conclusion

The Borovoy Coefficient does not replace Borovoy. It makes him operational. The diagnostic frame he carried across fifty years of Canadian civil-liberties practice — the intent-effect gap, the over-broad definition, the mesa-institutional drift of the commissions he helped build — turns out to be, when translated into the vocabulary of machine-learning alignment, a

precise technical claim. Governments and ML systems miss the mark through overlapping failure modes best catalogued in the four-variant Goodhart taxonomy. Their bureaucracies and neural networks alike are optimizers trained on proxies for under-specifiable goals, and they produce specification-gaming behaviour on distributional shifts from their formation conditions. Hennessy and Goodhart (2023) have shown that the two settings admit a common structural treatment; whether the resulting models are identical or merely kin is a question for a more technical paper than this one. The observation that the same structural techniques apply is enough to warrant an instrument.

If Borovoy were still in the field he would, I think, be unsurprised by this identification, even if he found the technology laughable. He spent his career reminding Canadians that noble purposes and broad drafting are a dangerous combination, and he spent it asking for parliamentary sunset clauses and hard-powered oversight rather than judicial discretion — the very reforms the alignment literature now calls *corrigibility* properties and proposes for systems that must remain under human control. “Obey the law but stick it to the government anyway,” he wrote in *Uncivil Obedience* (1991). Measurement, publication, and reform is the 2026 grammar of that disobedience.

I end with the line under which Borovoy chose to retire from the CCLA in 2009, and which is carved into the collective memory of Canadian civil-liberties work: “*forty years of raising hell without breaking the law.*” The Borovoy Coefficient is an attempt to raise hell legibly — to make the intent-effect gap something that can be measured, ranked, and reduced. This is a novel means by which constitutional marks ought not be missed.

Cite as: Michael Bryant, “The Borovoy Coefficient: A Civil-Liberties Measure of Legislative Alignment” (version 1.5, April 2026), justack.ai/borovoy. Released under the Mozilla Public License 2.0 as a composite measurement instrument with companion scoring methodology and open dashboard.

1. Christopher A Hennessy & Charles AE Goodhart, “Goodhart’s Law and Machine Learning: A Structural Perspective” (2023) 64:3 Int’l Econ Rev 1075, DOI 10.1111/iere.12633, open-access version at <https://eprints.lse.ac.uk/118656/>. ↩

2. Brian Christian, *The Alignment Problem: Machine Learning and Human Values* (New York: WW Norton & Company, 2020). ↩
3. Marian Botsford Fraser, *Acting for Freedom: Fifty Years of Civil Liberties in Canada* (Toronto: Second Story Press, 2014). ↩
4. My materials referenced “missing the mark” in a lecture delivered at the Massey College / University of Toronto privacy class (2020), and elaborated in a *Pandemic Litigation* lecture at McGill University Faculty of Law (9 November 2021). Borovoy’s own vocabulary favoured “too broad,” “vagueness,” and the intent-vs-effect distinction, as reflected in his testimony before the Special Senate Committee on Bill C-36 (24 October 2001): “*The crux of the issue, in our view, is that the bill is too broad. At the heart of that is the definition of ‘terrorist activity.’ Everything flows from that definition. ... It is important, however, that we distinguish intent from effect. Whether or not that was the intent of the bill, the risk is that that might be the effect of the bill.*” (cited *supra*.) The archery framing is the author’s analogical tribute to a great predecessor’s underlying idea. ↩
5. Biomedical misquotation: Gerald de Lacey, Carolyn Record & John Wade, “How Accurate Are Quotations and References in Medical Journals?” (1985) 291 *British Medical Journal* 884. Social-science replication: Open Science Collaboration, “Estimating the Reproducibility of Psychological Science” (2015) 349 *Science* aac4716 (36%); Colin F. Camerer et al., “Evaluating the Replicability of Social Science Experiments in *Nature* and *Science* Between 2010 and 2015” (2018) 2 *Nature Human Behaviour* 637 (62%). Journalism: Scott R. Maier, “Accuracy Matters: A Cross-Market Assessment of Newspaper Error and Credibility” (2005) 82 *Journalism & Mass Communication Quarterly* 533 (48%). Legal LLM hallucination: Matthew Dahl, Varun Magesh, Mirac Suzgun & Daniel E. Ho, “Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models” (2024) 16 *Journal of Legal Analysis* 64 (58-88% bare-model); Varun Magesh et al., “Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools,” *Journal of Empirical Legal Studies* (forthcoming 2025) (Lexis+ AI ~17%, Westlaw AI ~34%, Ask Practical Law ~17%). See also European Broadcasting Union & BBC, *News Integrity in AI Assistants* (October 2025); Richard Van Noorden, “More Than 10,000 Research Papers Were Retracted in 2023” *Nature* (12 December 2023); Daniele Fanelli, “How Many Scientists Fabricate and Falsify Research?” (2009) 4:5 *PLoS ONE* e5738. ↩
6. Sara Schroter et al., “What Errors Do Peer Reviewers Detect, and Does Training Improve Their Ability to Detect Them?” (2008) 101 *Journal of the Royal Society of Medicine* 507, <https://doi.org/10.1258/jrsm.2008.080062> (trained reviewers detected only 2.6 of 9 deliberately inserted major methodological errors at baseline; training produced small, non-durable improvements). ↩
7. Alan Borovoy, Testimony of the Canadian Civil Liberties Association, *Proceedings of the Special Senate Committee on the Subject-matter of Bill C-36*, 37th Parl, 1st Sess, Issue 3 (24 October 2001), <https://sencanada.ca/en/Content/Sen/committee/371/sm36/03eva-e>. See also Borovoy’s testimony before the House of Commons Standing Committee on Justice and Human Rights, 37th Parl, 1st Sess, Meeting 33 (24 October 2001). ↩
8. *Ibid.* ↩

9. Alan Borovoy, quoted in the public debate preceding the 2014 repeal of *Canadian Human Rights Act* s. 13; see also Alan Borovoy, interview with Caitlin Smith in *Freedom to Read* 2010, <https://www.freedomtoread.ca/articles/challenges-in-every-generation-alan-borovoy-in-conversation/> (“The law is so broad that you never know who or what might get nailed under it. There is no defence for truth, or reasonable belief in truth.”). ↩
10. *Canadian Human Rights Act*, RSC 1985, c H-6, s. 13 (as it read before repeal); repealed by *An Act to amend the Canadian Human Rights Act (protecting freedom)*, SC 2013, c 37, in force 26 June 2014. ↩
11. *R v Ndhlovu*, 2022 SCC 38. The majority reasons (Karakatsanis J.) rely on Crown-adduced evidence of recidivism rates to hold mandatory and lifetime sex-offender registration overbroad under s. 7, on the ground that automatic registration of a cohort of which 75-80% never reoffends is disconnected from the Act’s investigative and preventive purposes. The decision does not cite Goodhart’s Law, the machine-learning alignment literature, or the “Borovoy” vocabulary. It reasons in Canadian constitutional idiom. The present paper’s claim is that the Supreme Court’s reasoning in *Ndhlovu* maps structurally onto the regressional-Goodhart variant described in Manheim & Garrabrant (2018) — an instance in which a proxy has separated from its target under the pressure of enforcement. ↩
12. Norbert Wiener, “Some Moral and Technical Consequences of Automation” (1960) 131 *Science* 1355. ↩
13. Lon L. Fuller, *The Morality of Law* (Yale University Press, 1964), chap. 2, enumerating the eight ways law can fail “to be law on its own terms,” the eighth being “a failure of congruence between the rules as announced and their actual administration.” Fuller’s allegorical King Rex is, on a modern reading, a mesa-optimizer failing inner alignment. ↩
14. Evan Hubinger et al., *Risks from Learned Optimization in Advanced Machine Learning Systems*, arXiv:1906.01820 (2019). ↩
15. Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse & Scott Garrabrant, *Risks from Learned Optimization in Advanced Machine Learning Systems*, arXiv:1906.01820 (2019); Dario Amodei et al., *Concrete Problems in AI Safety*, arXiv:1606.06565 (2016). ↩
16. Amodei et al., *Concrete Problems in AI Safety*, *supra*; see also Pengfei Huang et al., “A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions,” arXiv:2311.05232 (2023) for a contemporary synthesis of divergence-management techniques. ↩
17. Charles A.E. Goodhart, “Problems of Monetary Management: The U.K. Experience” (1975), in *Papers in Monetary Economics*, vol I (Reserve Bank of Australia); Marilyn Strathern, ““Improving Ratings”: Audit in the British University System” (1997) 5 *European Review* 305. ↩
18. David Manheim & Scott Garrabrant, *Categorizing Variants of Goodhart’s Law*, arXiv:1803.04585 (2018). ↩
19. Victoria Krakovna et al., *Specification Gaming: The Flip Side of AI Ingenuity* (DeepMind Safety Research, 2020). ↩
20. Dario Amodei et al., *Concrete Problems in AI Safety*, arXiv:1606.06565 (2016). ↩

21. Michael K. Cohen, Marcus Hutter & Michael A. Osborne, “Advanced Artificial Agents Intervene in the Provision of Reward” (2022) 43 *AI Magazine* 282. ↩
22. James M. Buchanan & Gordon Tullock, *The Calculus of Consent: Logical Foundations of Constitutional Democracy* (University of Michigan Press, 1962); George J. Stigler, “The Theory of Economic Regulation” (1971) 2 *Bell Journal of Economics and Management Science* 3. ↩
23. James C. Scott, *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed* (Yale University Press, 1998). ↩
24. Donald T. Campbell, “Assessing the Impact of Planned Social Change” (1979) 2 *Evaluation and Program Planning* 67. ↩
25. *Ibid.* The paper’s own self-description is structural-econometric: it develops a model of penalized regression under adversarial covariate manipulation at known cost, and shows that the bias characteristics apply to both regulatory and machine-learning settings. The interpretive move of the present paper — that the common structural treatment warrants a common measurement instrument — is the author’s; the mathematical result is Hennessy and Goodhart’s. ↩
26. Jacob S. Hacker, “Privatizing Risk Without Privatizing the Welfare State: The Hidden Politics of Social Policy Retrenchment in the United States” (2004) 98 *American Political Science Review* 243. ↩
27. *Johnson v United States*, 576 US 591 (2015); *Sessions v Dimaya*, 584 US 148 (2018); *United States v Davis*, 139 S Ct 2319 (2019); *Romer v Evans*, 517 US 620 (1996). Reva Siegel’s “*The Rule of Love: Wife Beating as Prerogative and Privacy*” (1996) 105 Yale LJ 2117, which introduces the “preservation-through-transformation” framework, is the theoretical anchor of the US missing-the-mark tradition. ↩
28. *Bank Mellat v HM Treasury (No 2)* [2013] UKSC 39; Human Rights Act 1998 (UK), c 42, s. 4; *A v Secretary of State for the Home Department* [2004] UKHL 56. ↩
29. *West Virginia v Environmental Protection Agency*, 597 US 697 (2022); *Biden v Nebraska*, 600 US 477 (2023); *R (Miller) v Prime Minister* [2019] UKSC 41; *Salvesen v Riddell* [2013] UKSC 22; *AXA General Insurance Ltd v Lord Advocate* [2011] UKSC 46. ↩
30. *Canadian Human Rights Act*, RSC 1985, c H-6, s. 13 (as it stood before repeal); repealed by SC 2013, c 37 (in force 26 June 2014). See *supra* footnote on s. 13 repeal. ↩
31. Textual variations carry a range of implications, from trivial to catastrophic. Spell-check programs check textual variations like a (human) copy editor does for publications. A typo is an error and can be a significant error for some contexts (misspelling a prospective employer on a job application), trivial in others. Textual variations for encrypted identification can be catastrophic, for the US President approving a nuclear launch. Put another way, there is text and there is text. For the legal scholar, everything is text. Our discourse is through oral and written communications, captured in text. The implications of “textually-broken statutes,” therefore, deserves a library of commentary, though a section later will suffice (see *infra*). ↩
32. *Canada (Minister of Citizenship and Immigration) v Vavilov*, 2019 SCC 65, [2019] 4 SCR 653. ↩

33. *West Virginia v Environmental Protection Agency*, 597 US 697 (2022); *Biden v Nebraska*, 600 US 477 (2023); Human Rights Act 1998 (UK), c 42, s. 3; *Ghaidan v Godin-Mendoza* [2004] UKHL 30. ↩
34. *R v Oakes*, [1986] 1 SCR 103. ↩
35. *Buckley v Valeo*, 424 US 1 (1976); *Church of Lukumi Babalu Aye v Hialeah*, 508 US 520 (1993); *Reed v Town of Gilbert*, 576 US 155 (2015); *Bank Mellat v HM Treasury (No 2)* [2013] UKSC 39; *R (Daly) v Home Secretary* [2001] UKHL 26. ↩
36. *R v Heywood*, [1994] 3 SCR 761; *Canada (Attorney General) v Bedford*, 2013 SCC 72, [2013] 3 SCR 1101; *R v Ndhlovu*, 2022 SCC 38; *R v Nova Scotia Pharmaceutical Society*, [1992] 2 SCR 606. ↩
37. Hamish Stewart, “Overbreadth Revisited” (2024) 69:3 *McGill Law Journal* 247; see also Stewart, “Bedford and the Structure of Section 7” (2015) 60:3 *McGill Law Journal* 575. ↩
38. *Broadrick v Oklahoma*, 413 US 601 (1973); *United States v Stevens*, 559 US 460 (2010); *Johnson v United States*, 576 US 591 (2015); *Sessions v Dimaya*, 584 US 148 (2018); *United States v Davis*, 139 S Ct 2319 (2019); *A v Secretary of State for the Home Department* [2004] UKHL 56. ↩
39. *Youngberg v Romeo*, 457 US 307 (1982); *Washington v Glucksberg*, 521 US 702 (1997); *A v Secretary of State for the Home Department* [2004] UKHL 56; *R (Nicklinson) v Ministry of Justice* [2014] UKSC 38. ↩
40. *R v Le*, 2019 SCC 34, [2019] 2 SCR 692. ↩
41. *Washington v Davis*, 426 US 229 (1976); Reva Siegel, “*The Rule of Love: Wife Beating as Prerogative and Privacy*” (1996) 105 *Yale LJ* 2117 (introducing the “preservation-through-transformation” framework). ↩
42. *R (SC) v Secretary of State for Work and Pensions* [2021] UKSC 26; *R (Roberts) v Metropolitan Police Commissioner* [2015] UKSC 79. ↩
43. *Charkaoui v Canada (Citizenship and Immigration)*, 2007 SCC 9, [2007] 1 SCR 350, at para 53. ↩
44. *R v Ndhlovu*, 2022 SCC 38. See *supra* note on *Ndhlovu* and regressive Goodhart. ↩
45. *Johnson v United States*, 576 US 591 (2015); *Sessions v Dimaya*, 584 US 148 (2018); *United States v Davis*, 139 S Ct 2319 (2019). ↩
46. *Romer v Evans*, 517 US 620 (1996), at 632 (Kennedy J). ↩
47. *A v Secretary of State for the Home Department* [2004] UKHL 56; Human Rights Act 1998 (UK), c 42, s. 4. See also Anti-Terrorism, Crime and Security Act 2001 (UK), c 24, s. 23. ↩
48. *Salvesen v Riddell* [2013] UKSC 22; Scotland Act 1998 (UK), c 46, ss. 29(2)(d), 102; Agricultural Holdings (Scotland) Act 2003 (ASP 11), s. 72; European Convention on Human Rights, Protocol 1, Article 1. ↩

49. Robert K. Merton, “The Unanticipated Consequences of Purposive Social Action” (1936) 1 *American Sociological Review* 894. ↩
50. The Fullerian parallel is developed at greater length in Kristen Rundle, *Forms Liberate: Reclaiming the Jurisprudence of Lon L. Fuller* (Hart Publishing, 2012), and in Colleen Murphy, “Lon Fuller and the Moral Value of the Rule of Law” (2005) 24 *Law and Philosophy* 239. For the connection between Fuller’s eighth desideratum and contemporary AI-alignment vocabulary, see Alfred Tlaie, *Towards an Alignment Theory of Mind for Artificial Intelligence*, arXiv:2410.19749 (2024) — the only paper I have found that applies AI-alignment vocabulary explicitly to a regulatory instrument (there, the EU AI Act), though not to Fuller. ↩
51. Peter W. Hogg & Allison A. Bushell, “The Charter Dialogue Between Courts and Legislatures” (1997) 35 *Osgoode Hall Law Journal* 75. ↩
52. Stewart, “Overbreadth Revisited,” *supra*, at 247 (“[the strict formulation] relies on two untenable assumptions: (1) that a law is always an instrument for a purpose fully specifiable apart from legal order, and (2) that a law can be drafted so it never exceeds that purpose”). The present paper adopts Stewart’s diagnosis and reframes it through the AI-alignment literature as a case study of outer-alignment infeasibility. ↩
53. Colton Fehr, “Re-Thinking the Instrumental Rationality Principles of Fundamental Justice” (2020) 58:1 *Alta L Rev* 133. ↩
54. Peter W. Hogg, “The Brilliant Career of Section 7 of the Charter” (2012) 58 *Supreme Court Law Review* (2d) 195. ↩
55. Alana Klein, “The Arbitrariness in ‘Arbitrariness’” (2013) 63 *Supreme Court Law Review* (2d) 377. ↩
56. Reva B Siegel, “*The Rule of Love*: Wife Beating as Prerogative and Privacy” (1996) 105 *Yale LJ* 2117 (introducing the “preservation-through-transformation” framework). See also Siegel, “‘Why Equal Protection No Longer Protects’: The Evolving Forms of Status-Enforcing State Action” (1997) 49 *Stan L Rev* 1111; William J Stuntz, “The Pathological Politics of Criminal Law” (2001) 100 *Mich L Rev* 505; Samuel R Bagenstos, *Law and the Contradictions of the Disability Rights Movement* (New Haven: Yale University Press, 2009); Richard H Fallon Jr, “Strict Judicial Scrutiny” (2007) 54 *UCLA L Rev* 1267; Aziz Z Huq & Tom Ginsburg, “How to Lose a Constitutional Democracy” (2018) 65 *UCLA L Rev* 78. ↩
57. Hacker, *supra* note on policy drift. ↩
58. Neel Guha et al, “AI Regulation Has Its Own Alignment Problem: The Technical and Institutional Feasibility of Disclosure, Registration, Licensing, and Auditing” (2024) 92:6 *Geo Wash L Rev* 1473. ↩
59. Noam Kolt et al., *Legal Alignment for Safe and Ethical AI*, Oxford AI Governance Institute working paper (January 2026), arXiv:2601.04175. ↩
60. John Borrows, *Canada’s Indigenous Constitution* (University of Toronto Press, 2010); Val Napoleon, *Ayook: Gitksan Legal Order, Law, and Legal Theory* (PhD dissertation, University of Victoria, 2009); Aaron Mills,

"The Lifeworlds of Law: On Revitalizing Indigenous Legal Orders Today" (2016) 61:4 *McGill Law Journal* 847. ↩

61. Michael Chabon & Ayelet Waldman, eds., *Fight of the Century: Writers Reflect on 100 Years of Landmark ACLU Cases* (Avid Reader Press / Simon & Schuster, 2020); see also Ellis Cose, *Democracy, If We Can Keep It: The ACLU's 100-Year Fight for Rights in America* (The New Press, 2020). ↩
62. Marian Botsford Fraser, *Acting for Freedom: Fifty Years of Civil Liberties in Canada* (Toronto: Second Story Press, 2014). ↩
63. Liberty (formerly the National Council for Civil Liberties, founded 1934), <https://www.libertyhumanrights.org.uk/>; see *Liberty's Annual Report 2024-25* (London: Liberty, 2025) for the organization's current rights-defence agenda. ↩
64. OECD Observatory of Public Sector Innovation, "Rules as Code in Canada," 2024; New Zealand Service Innovation Lab, "Better Rules Approach," 2018. ↩
65. Mireille Hildebrandt, *Code-driven Law: Freezing the Future and Scaling the Past*, 2020; *The Cambridge Handbook of Private Law and Artificial Intelligence*, Cambridge University Press, 2024. ↩
66. Nyu Wang & Michael Yuan Tian, "'Intelligent Justice': human-centered considerations in China's legal AI transformation" (2023) 3:349 *AI & Ethics* Nyu Wang & Michael Yuan Tian, "'Intelligent Justice': human-centered considerations in China's legal AI transformation" (2023) 3:349 *AI & Ethics* ↩
67. See <http://justack.ai/blogs> for my efforts to provide empirical data or scholarly study on point. ↩